

## Definition of Project Management

Project management in the modern sense began in the early 1950s, although it has its roots much further back in the latter years of the 19th century. The need for project management was driven by businesses that realised the benefits of organising work around projects and the critical need to communicate and co-ordinate work across departments and professions. One of the first major uses of project management as we know it today was to manage the US space programme.

Project management is the discipline of planning, organizing, securing, managing, leading, and controlling of resources to achieve specific goals.

*The Project Management Institute's (2004)* define project management is the application of knowledge, skills, tools and techniques to project activities to meet project requirements.

According to *Chase and Aquilano*, Project Management can be defined as the planning, directing and controlling of resources (people, equipment, material) to meet technical, cost and time constraints of the project.

According to *B.B. Goel*, Project Management is an organized venture for managing projects that involves scientific application of modern tools and techniques in planning, implementing, monitoring, controlling and coordinating unique activities or tasks to produce desirable outputs in consonance with predetermined objectives within the constraints of time, cost, quantity and quality.

*Richard P. Olsen*, in his article "Can Project Management Be Defined?" defined project management as "the application of a collection of tools and techniques...to direct the use of diverse resources toward the accomplishment of a unique, complex, one-time task within time, cost, and quality constraints. Each task requires a particular mix of these tools and techniques structured to fit the task environment and life cycle (from conception to completion) of the task."

Project management is often summarised in a triangle (see Figure 1). The three most important factors are time, cost and scope, commonly called the triple constraint. These form the vertices with quality as a central theme.



*Figure 1: The triple constraint*

- Projects must be within cost.
- Projects must be delivered on time.
- Projects must be within scope.
- Projects must meet customer quality requirements.

More recently, this has given way to a project management diamond, with cost, time, scope and quality the four vertices and customer expectations as a central theme (*see* Figure 2). No two customer expectations are the same so you must ask what their expectations are.



*Figure 2: The project management diamond*

## Project Management Basics

No matter what the type of project, project management typically follows the same pattern:

### 1. Defining the Project

In this stage the project manager defines what the project is and what the users hope to achieve by undertaking the project. This phase also includes a list of project deliverables, the outcome of a specific set of activities. The project manager works with the business sponsor or manager who wants to have the project implemented and other stakeholders -- those who have a vested interest in the outcome of the project.

### 2. Planning the Project

Define all project activities. In this stage, the project manager lists all activities or tasks, how the tasks are related, how long each task will take, and how each task is tied to a specific deadline. This phase also allows the project manager to define relationships between tasks, so that, for example, if one task is x number of days late, the project tasks related to it will also reflect a comparable delay. Likewise, the project manager can set milestones, dates by which important aspects of the project need to be met.

Define requirements for completing the project. In this stage, the project manager identifies how many people (often referred to as "resources") and how much expense ("cost") is involved in the project, as well as any other requirements that are necessary for completing the project. The project manager will also need to manage assumptions and risks related to the project. The project manager will also want to identify project constraints. Constraints typically relate to schedule, resources, budget, and scope. A change in one constraint will typically affect the other constraints. For example, a budget constraint may affect the number of people who can work on the project, thereby imposing a resource constraint. Likewise, if additional features are added as part of project scope, that could affect scheduling, resources, and budget.

### **3. Executing the Project**

Build the project team. In this phase, the project manager knows how many resources and how much budget he or she has to work with for the project. The project manager then assigns those resources and allocates budget to various tasks in the project. Now the work of the project begins.

### **4. Controlling the Project**

The project manager is in charge of updating the project plans to reflect actual time elapsed for each task. By keeping up with the details of progress, the project manager is able to understand how well the project is progressing overall.

### **5. Closure of the Project**

In this stage, the project manager and business owner pull together the project team and those who have an interest in the outcome of the project (stakeholders) to analyze the final outcome of the project.

## **Importance of Project Management**

Project management has emerged as a crucial factor that determines the success of an organization. Whether it is a question of facing an economic crisis or generating large turnover, it plays a pivotal role in the growth of a firm. Since time immemorial, man has incorporated the idea of project management in the construction of monuments and various other architecture. However, since the early 1950s, it started to be pursued as a major discipline in the corporate companies. It forms the backbone in the fields of civil engineering, defense, aerospace engineering and infrastructure development.

Henry Gantt, known as the father of planning and control techniques, and Henri Fayol, famous for his 5-point management theory on project management, are regarded as the two stalwarts in this field. Companies recruit project managers that lead the team in order to timely achieve the targets set by the clients and customers. In management terminology, the duration in which an entire project is carried out, from its inception till the end, is called the project life cycle. The project life cycle can consist of one or multiple approaches. The management approaches are the various methodologies that the team can decide to follow so that the target is achieved in an allotted time. The most basic analysis of the project is called the initiation stage of a project. After this stage, designing or planning is done to streamline the task and make a flexible strategy to complete the task. Then the execution phase takes the center stage, and in this phase, the project is practically tackled with all its pros and cons. The monitoring and controlling of the project is done with the execution stage to check the possible threats and keep an eye on the performance of the team. Once the project is about to finish, it goes into the closing phase, where all the activities are finalized.

Some of the prime advantages of having a good project management team for a company are as follows.

#### **1. Excellent product quality**

Consumers generally look for low cost and high quality, while purchasing a product. Maintaining a high standard of excellence in developing quality products earns the company goodwill amongst

its customers. The plans the allocated budget, resources and testing methods that keep the pace of production high, both qualitatively and quantitatively. The project management teams can also undertake six sigma training programs that enhance the quality of the products. So, we can say that project managers indirectly contribute a lot in ensuring that the quality of products is consistently high. In the recent years, a large number of standard corporate training programs have been designed to help project managers understand the intricacies of quality, control and management.

## **2. Adequate communication**

Improper communication among employees can lead to misunderstandings and negatively impact the performance of the firm. A project manager can be a bridge among the diversified branches of project undertaking. And why only employees? Stakeholders also form a part of the company. They prefer investing in those companies that deliver projects on time and keep them informed about updates and progress of the projects. If a client is satisfied with the performance of the firm, it is likely that it will return with much bigger projects, not to mention huge investments. A project leader can hold meetings on a daily, weekly or monthly basis and can make sure that everyone is aware about the project plan and his/her responsibilities, both as an individual and as a team. With the help of human resource management department, project managers can be more effective in communicating the expectations of the clients.

## **3. Reducing risks**

The probability of getting hit by an unwanted or unexpected event has increased manifold in today's competitive business environment. The project management team can identify the potential risks, take their time to rectify them and help the company save valuable resources. In the case of a worst crisis, this team can opt for change management method to attain the desired goals. Team work is a must, when it comes to visualizing the dangers ahead. Risk management principles can be applied by the project managers to eliminate risks to a larger extent.

## **4. Strategic objectives and goals**

Strategic goals are the blueprint of the task undertaken by a company. For instance, a software company aims to prepare software and related programming codes, whereas an infrastructure company has a target of constructing dams, bridges and other construction works. A project management team helps the company in achieving the strategic goals, as it streamlines the task of a company in taking many important decisions. Strategic planning and strategic thinking are vital management tools for a project management team.

Once the task is allotted, the project team is responsible for the goal to be finished in the dedicated time. Innovation is an area in which the project team can invest more and come out with new ideas that can increase the profit margin and also reputation of the firm. Human resource, financial planning, corporate social responsibility and physical resources are other facets of strategic goals.

## **5. Crisis Management**

Crisis in a project can occur because of several reasons like recession, labor strikes, paucity of funds and natural calamities, to just name a few. In such a situation, it is essential for project managers to display maturity and profound administrative and managerial skills to handle the

situation. Sometimes, abandoned project and tasks are completed successfully, under the able guidance of project managers. The power inherent in the abilities of an accomplished project manager makes a huge difference in the outcome of the project.

It is an essential segment in every organization. Be it the small scale enterprises or corporate giants, project management has the power to transform the market standing of a company and help it soar high in the sky.

## **Process of Project management**

The project management process typically includes four key phases: initiating the project, planning the project, executing the project, and closing the project. An outline of each phase is provided below.

### **1. Initiating the Project**

The project management techniques related to the project initiation phase include:

1. *Establishing the project initiation team.*
2. *Establishing a relationship with the customer.*
3. *Establishing the project initiation plan..*
4. *Establishing management procedures.*
5. *Establishing the project management environment and workbook.*

### **2. Planning the Project**

The project management techniques related to the project planning phase include:

1. *Describing project scope, alternatives, and feasibility.* The understanding of the content and complexity of the project. Some relevant questions that should be answered include:
  - What problem/opportunity does the project address?
  - What results are to be achieved?
  - What needs to be done?
  - How will success be measured?
  - How will we know when we are finished?
2. *Divide the project into tasks.*
3. *Estimating resources and creating a resource plan.*
4. *Developing a preliminary schedule.*
5. *Developing a communication plan.*
6. *Determining project standards and procedures.*
7. *Identifying and assessing risk.*
8. *Creating a preliminary budget.*
9. *Developing a statement of work.*
10. *Setting a baseline project plan.*

### **3. Executing the Project**

The project management techniques related to the project execution phase include:

1. *Executing the baseline project plan.*
2. *Monitoring project progress against the baseline project plan.*
3. *Managing changes to the baseline project plan.*
4. *Maintaining the project workbook.*
5. *Communicating the project status.*

#### **4. Closing down the Project**

The project management techniques related to the project closedown phase include:

1. *Closing down the project.*
2. *Conducting post project reviews.*
3. *Closing the customer contract.*

#### **Success factors in Project Management**

Baker, Murphy and Fisher have conducted a research survey of 650 projects to identify the success and failure factors. While the general perception of successful projects includes fulfillment of the time and cost schedule and technical specification, their research finding puts foremost emphasis on final project performance, achievement of the project's mission and perceptions of different stake holders of a project. They conclude that -"In the long run, what really matters is whether the parties associated with, and affected by a project are satisfied." Some of the prerequisites of project success may be identified as under:

- A thorough feasibility study which does not overlook any significant element affecting the project.
- A capable project manager and a good team around him, with involvement right through all four phases of project.
- An adequate organisation with proper systems of communication, feedback and control mechanisms.
- Adequate understanding of mutual roles and responsibilities of all interacting agencies with clear understanding of demarcations, dependencies and complementarities.
- Quick identification and resolution of conflicts that is nonetheless inevitable.
- Adequate and timely funding.
- Certain degree of insulation to front line project managers from undue impacts from the environment.
- Regular updation and appropriate actions to continuously correct the deviations, till the end of the project.

The above list is not at all exhaustive, but includes some of the more important elements for a successful project. We hope you would enjoy reading the subsequent units that would unfold this subject to you, in greater details.